Expected value and variance of trace function

For random variables $X \in \mathbb{R}^h$, and a positive semi-definite matrix A: Is there a simplified expression for the expected value, $\mathbb{E}[Tr(X^T AX)]$ and variance, $Var[Tr(X^T AX)]$? Please note that A is not a random variable.

variance mathematical-statistics expected-value random-matrix

edited Aug 16 '12 at 20:59
mbq
19.5k 10 57 111

asked Aug 16 '12 at 20:33
hotchpotch
1,546 10 23

1 Answer

Note that $\mathrm{Tr}(X^T AX) = X^T AX = \mathrm{Tr}(AXX^T)$ so that
$\mathrm{E}(X^T AX) = \mathrm{E}(\mathrm{Tr}(AXX^T)) = \mathrm{Tr}\mathrm{E}AXX^T = \mathrm{Tr}(A\mathrm{E}(XX^T))$.

Here we have used that the trace of a product are invariant under cyclical permutations of the factors, and that the trace is a linear operator, so commutes with expectation. The variance is a much more involved computation, which also need some higher moments of X. That calculation can be found in Seber: "Linear Regression Analysis" (Wiley)

answered Aug 16 '12 at 20:51
kjetil b halvorsen
20.9k 9 61 149